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MSc THESIS

MicroRNA target prediction with novel machine learning methods beyond quadratic attention

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ΔΙΠΛΩΜΑΤΙΚΗ ΕΡΓΑΣΙΑ

Πρόβλεψη στόχων μικροRNA με νέες μεθόδους μηχανικής μάθησης πέρα από τις τετραγωνικές αρχιτεκτονικές

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Οκτώβριος 2025

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ABSTRACT

MicroRNAs are small, non-coding RNA molecules that partake in a variety of regulatory mechanisms, with the most well-studied being their role in the suppresion of translation of mRNAs via binding to sequences called MicroRNA Response Elements (MREs). They are thus implicated in a wide range of mechanisms in physiology and their disregulation can be decisive for disease states. In the current age of multi-omics analyses and personalized medicine, miRNAs are discovered via high-throughput sequencing experiments such as small RNA-seq and their regulatory roles can be studied via experiments such as AGO-eCLIP and AGO-CLASH that rely on the quantification of miRNA-mRNA hybrids. Given that a miRNA can target several different mRNA molecules, it is often complex to discern the regulatory effect of many microRNAs from a single high-throughput sequencing experiment which necessitates the use of advanced computational analyses for this purpose. To that end, several machine learning and deep learning models have emerged with the goal of miRNA target prediction. Such models aim either to predict valid miRNA-MRE pairs, or make predictions at the gene level, either finding possible MREs within the entire gene/mRNA sequence or using this information to predict changes in gene expression with the integration data from additional high-throughput sequencing experiments regarding the transriptome. The goal of this thesis is to develop a machine/deep-learning model that, given pairs of microRNAs and MREs predicts which ones would interact. For this purpose, AGO2-eCLIP data were utilized for training and evaluation of several models leading to the development of an weighted ensemble of machine learning models that is able to discern valid from invalid such pairs with an Average Precision score comparable to other state of the art models trained on the same task. The output of the model also allows for advanced interpretability via providing scores for the contribution of each feature to the final prediction.

SUBJECT AREA: Thesis LaTeX template

KEYWORDS: list, of, keywords

ΠΕΡΙΛΗΨΗ

Τα microRNAs είναι μικρά, μη κωδικοποιούντα μόρια RNA που συμμετέχουν σε διάφορους ρυθμιστικούς μηχανισμούς. Σε αυτή την εργασία, αναπτύσσουμε μοντέλα μηχανικής μάθησης για την πρόβλεψη αλληλεπιδράσεων μεταξύ microRNAs και στόχων mRNA. Χρησιμοποιώντας δεδομένα AGO2-eCLIP, δημιουργήσαμε ένα weighted ensemble μοντέλο που επιτυγχάνει υψηλή απόδοση σε μια ισορροπημένη κατανομή των δεδομένων κατά οικογένεια microRNA.

ΘΕΜΑΤΙΚΗ ΠΕΡΙΟΧΗ: Πρότυπο διπλωματικής εργασίας LaTeX

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: μικροRNA, πρόβλεψη στόχων, μηχανική μάθηση

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PREFACE

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1. INTRODUCTION

1.1 MicroRNAs

MicroRNAs are the most well characterized class of non-coding RNA molecules, most commonly between 18-22 nucleotides long, that act mainly to regulate the post-transcriptional landscape of the cell, functioning as small guide RNAs for proteins of the Argonaute (AGO) family. Typically, miRNAs bind to the target mRNA sequence via their seed region which covers positions 2-8 in their sequence. Perfect complementarity leads to AGO mediated cleavage of the target with the Argonaute proteins functioning as endonucleases. Consequently, the translation of the target mRNA molecule halts and it is further targeted by other cellular RNA nucleases that lead to its complete degradation. In that manner, they are able to function in a vast variety of regulatory networks that control cellular behaviour. MicroRNAs are named based on the miRBase scheme whereby a 3-letter code denotes the species the miRNA was discovered in (e.g hsa: Homo sapiens), followed by miR in the case of mature sequences (or mir for precursor ones) and finally a number that conveys when the molecule was discovered/published with respect to the other entries (AMBROS *et al.* 2003; Kozomara & Griffiths-Jones 2011). As a rule, miRNAs from different organisms that are orthologous share the same name after the species identifier, whereas paralogous sequences have a single-letter suffix that distinguishes them and hairpin loci that give rise to the same mature miRNA are specified by a single-digit suffix. Finally, in cases where two different molecules are excised from the same precursor, a 5p or 3p suffix is used. It is also worth noting that with the advent of advanced high-throughput sequencing techniques, it has become plausible to detect isoforms of miRNAs. Such molecules differ from the canonical miRNA sequence either in terms of length of their 5' and/or 3' ends or because their sequence has been modified via processes like A-to-I or C-to-U editing. These modifications subsequently change how they are loaded to AGO and how they interact with target MREs (Tomasello *et al.* 2021).

1.1.1 Biogenesis

The transcription of a miRNA gene is performed by RNA Polymerase II or RNA Polymerase III and leads to the production of a precursor called pri-miRNA which is longer in length than the mature molecule and has the structure of a stem-loop with single-stranded overhangs extending from its 3' and 5' ends. It is necessary that a stem of a pri-miRNA molecule is of specific size, its loop is at least ten base pairs long and its ends can produce single-stranded RNAs, so that it can be processed into a mature miRNA (Zeng & Cullen 2003; Zhang & Zeng 2010). The maturation process starts with a protein complex comprising of an endoribonuclease called Drosha and a protein called DGRC8. DGRC8 is a protein that binds double-stranded RNA, in this case the stem of the pri-miRNA, stabilizing its interaction with Drosha. The cleavage of the pri-miRNA by Drosha is also guided by DGRC8 which recognizes the region 11 base pairs downstream from its binding site.

The product is pre-miRNA, a molecule that is 60-70 base pairs long and that is exported from the nucleus by exportin 5. An alternative origin for miRNA genes is introns of protein coding genes, such genes are also called mirtrons. Due to the fact that the pri-miRNA molecules originate from introns,they do not have the appropriate structure needed in order to be processed by Drosha and DGRC8. A protein called DRB1 solves the lariat structure of intronic RNA and pre-miRNA is the immediate product of this process. Another example of Non-canonical miRNA biogenesis includes 7-methyl Guanosine (m7G)-capped pre-miRNA, which are exported to the cytoplasm via Exportin 1 without being cleaved by Drosha first. All pre-miRNAs are recognized by a type 3 RNAse called DICER and TRBP which is a protein that binds the stem of the pre-miRNA. DICER can then cleave the pre-miRNA molecule into two leading to products of 22 base pairs long, one of which will be loaded into the RISC complex that is comprised of DICER, TRBP, AGO, PACT, and TNRC6A-C (O'Brien *et al.* 2018).

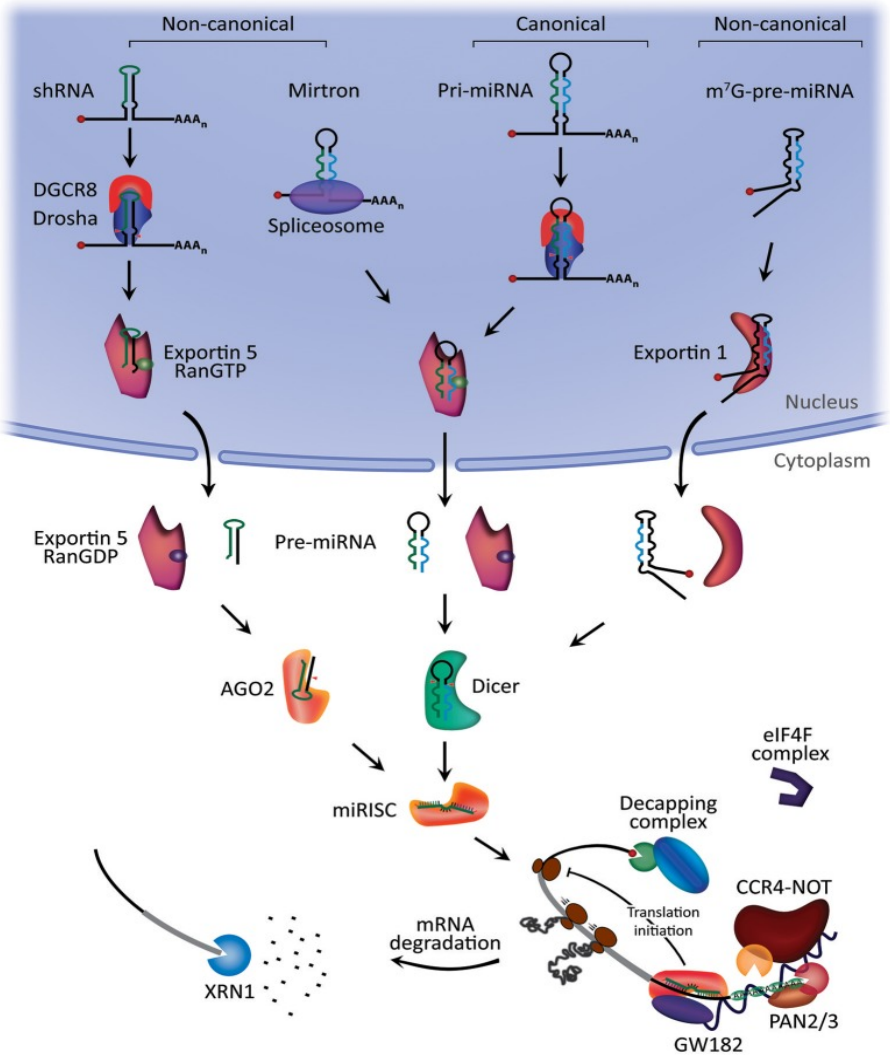

Figure 1.1: The steps of miRNA biogenesis. (O'Brien *et al.* 2018)

1.1.2 Target Binding

The strand that is loaded is to be used as a guide RNA for RISC, via hybridizing with the mRNA. The dsRNA is mainly formed between the seed region of the miRNA (nucleotides 1-8) and the 3'UTR of the mRNA, with additional, downstream base pairing aiding the interaction. Given perfect complementarity, the endonuclease activity of AGO2 is induced and the mRNA is cleaved. At the same time, proteins of the GW182 complex are recruited in order to act as scaffolds for necessary effector proteins such as PAN2-PAN3 and CCR4-NOT complexes that function as Poly-A Deadenylases that remove the protective 3' polyA tail, given the interaction between tryptophan from W repeats of the GW182 complex and the PABC complex on the mRNA has been established. Finally, removal of the 5'cap of the mRNA from DCP2 takes place followed by 5'-3' degradation from the exonuclease XRN1.

Most of the miRNA:MRE complexes in the cell are not perfectly complementary and additional 3' base pairing plays an important role for their stabilization. The regulatory efficiency scales with the number of seed matches according to the following order (Grimson *et al.* 2007):

1. 8mer: 8 matches within the seed region (1-8)
2. 7mer m8: 7 matches within the seed region (2-8)
3. 7mer A1: 7 matches within the seed region with the first nucleotide being an Adenine (1-7)
4. 6mer: 6 matches within the seed region (2-7)
5. 6mer offset: 6 matches within the seed region (3-8)
6. 5mer: 5 matches within the seed region (2-6)

As mentioned above, additional base pairing can enhance the stability of the complex and is classified as supplementary binding if it does so without being necessary for the regulatory role of the miRNA, or as compensatory if it compensates for mismatches within the seed region. Finally, there is a seedless mode of pairing for miRNAs that has been described in C.elegans which likely occurs due to AU rich sequences in the seed region. Regulatory efficiency is also affected by relationships between different binding sites and binding events within the same mRNA molecule. Adjacent sites for different miRNAs and repetitive sites for the same miRNA can both function in a cooperative manner. Functional cooperativity refers to the event whereby many different binding events have a cumulative effect, whereas binding cooperativity to the physical interaction of different RISC complexes supporting each other in the binding of their target (Diener *et al.* 2024).

1.1.3 Additional mechanisms of action

Binding of miRNAs on MREs can also increase the efficiency of translation instead of suppress it. A protein called FXR1 is necessary for this phenomenon, which in combination with RISC may bind specific sequence motifs (such as AU Rich Elements, AREs) or other sequences in the 5'UTR leading to activation of translation. MicroRNAs can also regulate gene expression via utilizing other, albeit less studied, mechanisms. For example, RISC complexes can be transported in the nucleus in order to exert regulation on transcription. In order for this to occur, they have to be imported from the cytoplasm via importin 8, importin α/β or XPO1. Within the nucleus, these complexes can bind to DNA regulatory elements affecting the binding of transcription factors and of other regulatory proteins (such as chromatin remodelling complexes) on them, thus having an impact on the transcription of related genes (Kim *et al.* 2008; Chaluvally-Raghavan *et al.* 2016; Xiao *et al.* 2017; Guo *et al.* 2021). A noteworthy example is that of miR-30b-5p which upon starvation, translocates to the nucleus in order to bind regulatory elements known as CLEAR elements that are known to bind the TFEB transcription factor. In that manner, it suppresses the transcription of these genes and inhibits lysosomal biogenesis and the autophagic flux (Guo *et al.* 2021). Additionally, RISC has been shown to be able to bind to mRNAs within the nucleus and induce their degradation while AGO and Drosha also participate in the splicing of premature mRNAs (O'Brien *et al.* 2018).

1.1.4 Roles in Physiology and Disease

Given their diversity regarding both sequence and mechanisms of action, miRNAs partake in many physiological and pathological states. Due to their capability of regulating multiple genes in a variety of different manners, they are important components in developmental processes that require the fine-tuning of gene expression at precise time intervals, with a prominent example being that of miR-196. Anterioposterior polarity is mediated by the SHH protein in forelimb and hindlimb buds and its expression requires Retinoic Acid (RA) signalling induced Hoxb8 expression. Hoxb8 is a transcription factor protein that is upregulated as an immediate-early response to ectopic RA administration to forelimb but not to hindlimb buds due to a specific miRNA, miR-196, that inhibits the translation of Hoxb8 in a tissue specific manner (Hornstein *et al.* 2005). MicroRNAs are also able to regulate protein synthesis spatiotemporally within the cell, localizing to specific subcellular compartments before exerting their effects (Corbin *et al.* 2009).

Deregulation of miRNA expression can lead to or deteriorate a big number of disease states ranging from congenital defects to autoimmune diseases, neurological disorders and many types of cancer. For instance, Adam *et al.* 2009 showed that miR-200 family expression in UMUC3 cells lead to increase in E-Cadherin expression and elevated sensitivity to EGFR inhibitors. MiRNAs can also be excreted from one cell to be taken in by a target via exosomes. Lin *et al.* 2019 demonstrated that miR-21 is excreted from bladder cancer cells via exosomes that are taken in by macrophages in order to activate the PI3K/AKT pathway and promote cancer progression by polarizing them towards the M2

phenotype. In addition, miRNAs can be detected in fluid samples from patients and can thus be used as biomarkers for specific diseases. For instance, Sapre *et al.* 2016 showed that a combination of 6 miRNAs from urine are able to predict Urothelial Carcinoma of the Bladder (UCB) patients with a high AUC score (0.85). Finally, miRNAs can be used as targets for therapeutics that either induce or reduce their expression. An example is miR-222 which has elevated expression levels in cardiomyocytes after specific types of exercise as well as following exercise rehabilitation after heart failure. This elevated expression leads to targeting of p27, HIPK-1, HIPK-2, and HMBOX1 which promotes cardiomyocyte hypertrophy, proliferation, and survival (Ding *et al.* 2017).

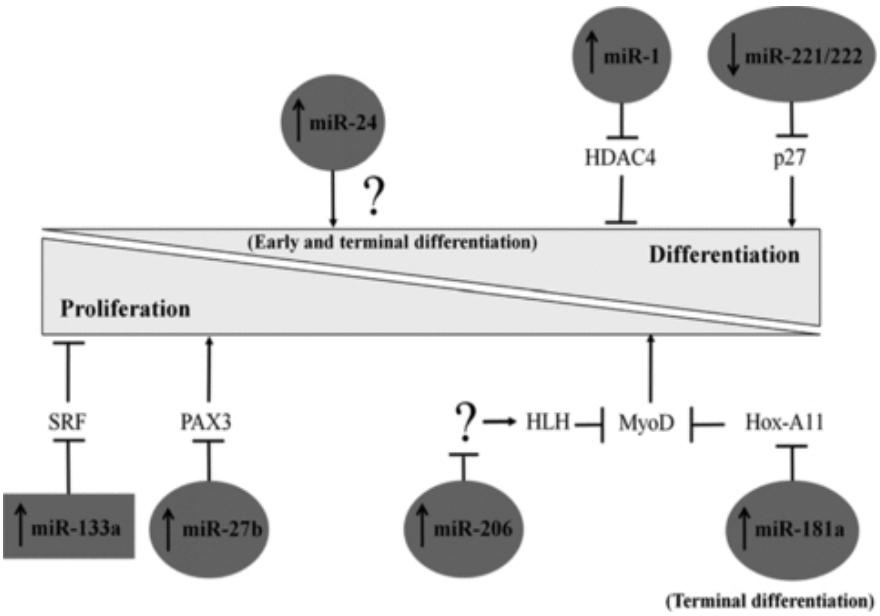

Figure 1.2: MiRNAs regulating muscle differentiation. (Güller & Russell 2010)

1.1.5 High-throughput sequencing methods for miRNA research

In the era of multiomics studies, it has become prevalent to study the totality of the miRNA part of the transcriptome as well as their targets via the use of high-throughput sequencing techniques. Such techniques allow for the detection and quantification of all the miRNA molecules in a tissue as well as of the complexes that these form with MREs. The most widely used one is small RNA-seq which includes isolation of RNA, cDNA library construction and sequencing. During RNA isolation, the total RNA might be extracted from a tissue, optionally followed by enrichment methods such as length separation or binding of small RNAs to to specific proteins. Library construction involves a reverse transcription step followed by ligation of adapters and/or polyadenylation (Benesova *et al.* 2021). The following image depicts the double ligation small RNA-seq protocol:

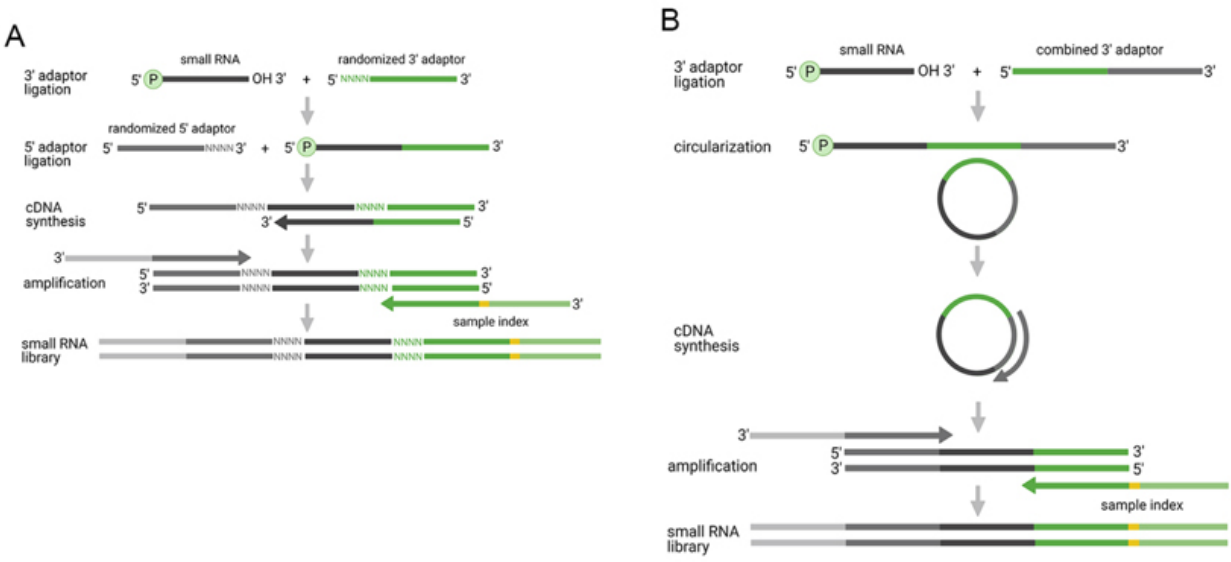

Figure 1.3: Double ligation small RNA-seq. (Benesova *et al.* 2021)

Types of experiments that capture and quantify miRNA:MRE complexes are used to study the binding sites of miRNAs and include AGO-eCLIP and AGO-CLASH. The latter involves the following steps (A *et al.* 2013):

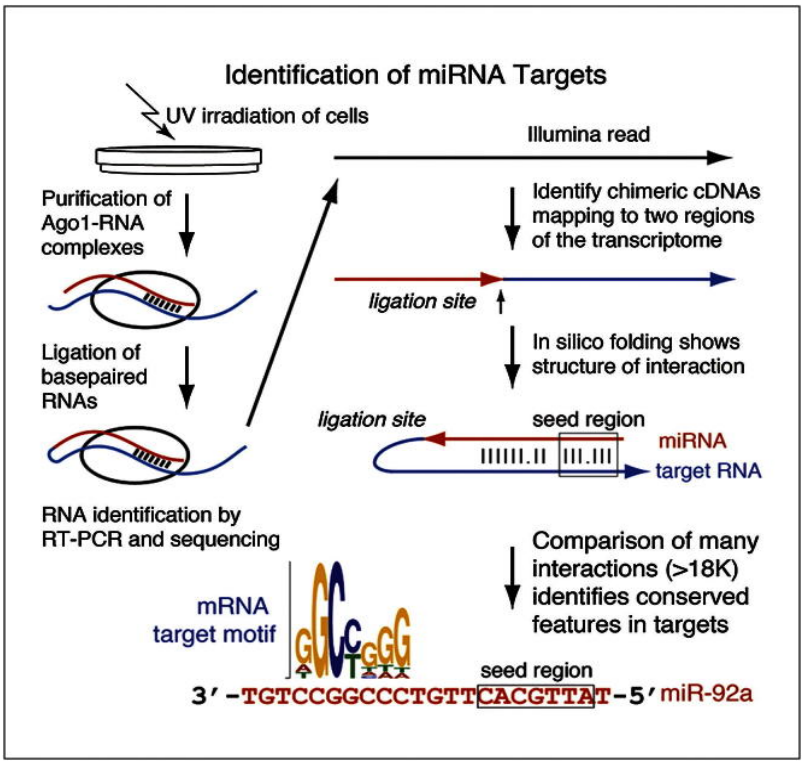

Figure 1.4: AGO-CLASH. (A *et al.* 2013)

Before cross-linking via UV irradiation, AGO expression is induced via the use of Doxicyclin, where the concentration is monitored in order for AGO to be close to endogenous levels. Despite that, differences between endogenous expression do occur and with them, the risk of bias regarding specific interactions. After cross-linking, hydrolysis of interacting RNA molecules takes place followed by ligation of the hybridized miRNA:MRE duplex which creates hybrid molecules to be reverse transcribed, amplified and sequenced. AGO-eCLIP follows a different procedure that does not require modifications to the endogenous expression. During AGO-eCLIP, the RISC:MRE complex is cross-linked using UV irradiation and the RNA is fragmented in the same manner as with AGO-CLASH, then immunoprecipitation, using antibodies that bind the AGO protein, follows and ultimately ligation of the miRNA 5'-P to the mRNA's 3'-OH group along with adapter ligation to the mRNA's 5' end (Manakov *et al.* 2022). Due to the immunoprecipitation step, no exogenous induction of expression is needed, eliminating the aforementioned bias and providing an image that is more physiologically relevant.

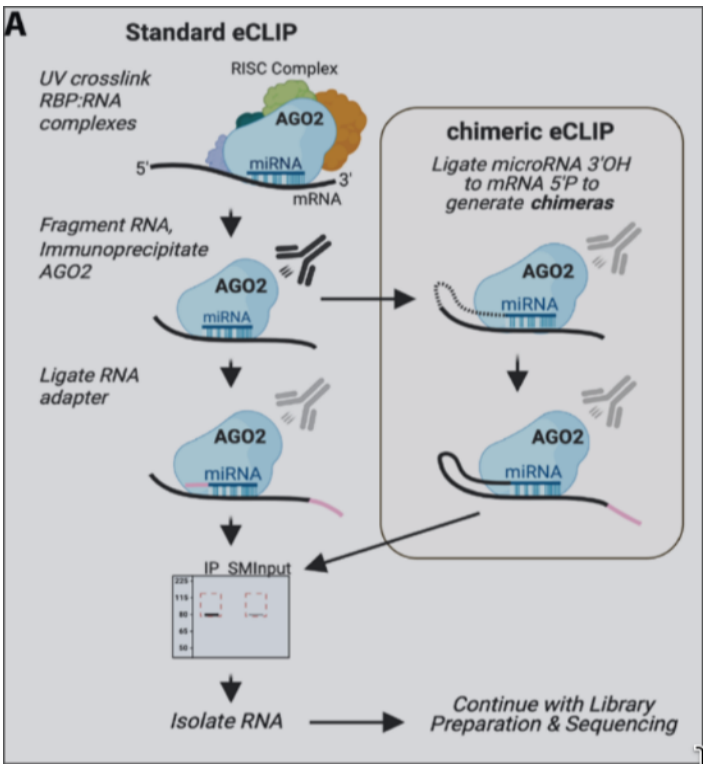

Figure 1.5: AGO-eCLIP. (Manakov *et al.* 2022)

There is also a family of techniques used to quantify the regulatory effect of miRNAs on their targets, which includes Degradome Sequencing, Parallel Analysis of RNA Ends and Genome-Wide Mapping of Uncapped and Cleaved Transcripts (Gregory *et al.* 2008; Addo-Quaye *et al.* 2008; German *et al.* 2009). In these experiments, the polyA transcripts are isolated from the tissue of interest, then ligation of the first adapter occurs only on transcripts with 5' monophosphate (which are the ones that have lost their 5'm7G cap due to miRNA mediated decay), followed by reverse transcription, 3' adapter ligation, PCR and

sequencing. The result is depicted as a peak on an mRNA transcript where the x-axis shows the sequence the miRNA targeted and the y-axis the frequency of 5' ends found at this position.

High-throughput sequencing techniques produce enormous amounts of data regardless of their end goal. Commonly, the immediate result from the sequencing instrument involves files that are called FASTQ files which include entries of all the sequenced molecules along with a quality score that conveys the reliability of the base that was sequenced at a given position. Each entry of a FASTQ file follows the format below:

1. 1st Line: @ character followed by an identifier for the entry
2. 2nd Line: Raw nucleotide seq
3. 3rd Line: + character optionally followed by the sequence identifier again
4. 4th Line: quality values for the sequence

It is worth noting that there is a number of different sequencing instruments that can be used for the same protocol but utilize different biochemistry principles in order to perform sequencing at such large scales. Due to these differences, the end result will be different even if one were to sequence the exact same samples and as such specific methods have been developed to mitigate this issue and ensure reproducibility across studies. Such methods are necessary when one wants to combine results from a number of different studies that have performed these experiments and the process called Batch Effect Correction. Finally, even within the same experiment it is heavily suggested that there are more than one replicates for the same sample in order to account for technical variation in the measurements.

Depending on the sequencing protocol and the end goal, these files are processed following different pipelines which include the usage of a series of computational tools. A common step regardless of the protocol is the Quality Control step (QC) in which various plots are produced in order to examine the integrity of the sequenced molecules based on the quality scores themselves as well as knowledge regarding the expected nucleotide and length distributions. Depending on the results, the next step involves filtering based on specific thresholds and optional removal of the adapter sequences followed by another QC step to ensure integrity has been maintained.

The next step is to map the sequences in the filtered file on the human genome/transcriptome and it is the first source of heterogeneity pertaining to such pipelines. Tools that perform this mapping are called Aligners and there is a big number of such tools each of which excels in specific circumstances. For small RNA-seq, two aligners that are commonly used are Bowtie2 and BWA-MEM (Langmead *et al.* 2009; Li 2013), that achieve high speed alignment of short reads on genomic sequences. The choice of the alignment tool as well as that of its alignment parameters are vital for the quality of the end result which necessitates their inclusion in the pipeline for reproducibility. A vital aspect of this step is the genome version that is used where the reads are to be aligned, as different versions lead

to differences in results. Most studies use hg38 which is considered the more up to date and complete version, although with deeper and more accurate sequencing more recent versions like the Telomere to Telomere version have emerged which might come to replace hg38, especially for the analysis of repetitive regions such as centromeres, telomeres and ribosomal array sequences. For AGO-eCLIP and AGO-CLASH, a typical aligner would not suffice as the reads that are sequenced are chimeric molecules, consisting partly of miRNA and of mRNA fragments. Thus, specific tools have been developed that can appropriately align each part, with an example being Hyb (Travis *et al.* 2014). Hyb begins by aligning the reads either to the genome or to a known database of molecules (ideally mature mRNAs) in order to identify reads that are non-contiguously aligned and then utilizes folding algorithms to annotate the recovered duplexes.The end result is a list of RNA:RNA hybrids.

1.2 miRNA Target Prediction Algorithms

Several methods have been developed over the years that utilize the data produced by the aforementioned experiments in order to achieve several goals related to elucidating miRNA biology. In general, predictive algorithms are separated in terms of the type of algorithms they use in order to make the prediction as well as the prediction task itself.

- Algorithms
  - Heuristic
  - Physics Based
  - Machine/Deep Learning
- Task
  - Site Level
  - Gene/Functional Level
  - Pathway Level

1.2.1 Site Level predictors

In terms of algorithms, a tool is characterized as Heuristic if it makes predictions using preset rules that are based on already discovered interactions (Lewis, Shih, *et al.* 2003; Lewis, Burge, *et al.* 2005). The first release of TargetScan was heuristics based, as it searched for perfect seed complementarity in the 3'UTR of each target provided, extended it to include as many matches as possible and then used software like RNAFold to check for additional complementarity in the miRNA 3'UTR with the next 35 nucleotide of the mRNA (Hofacker *et al.* 1994) and assigned a Z score that is the basis for the final predictions.

Such tools are considered the simplest form of prediction tools but function as a good baseline for comparison with more complex models.

Physics based tools may disregard the biology behind the task and focus on expressing the problem as an energy minimization one when it comes to MRE prediction although more recent ones aim to strike a balance between using some heuristics for narrowing down some predictions in combination with energy minimization. A fine example of this is IntaRNA (Busch *et al.* 2008; Mann *et al.* 2017; Gelhausen *et al.* 2019; Wright *et al.* 2014; Miladi *et al.* 2019; Raden *et al.* 2020) which is a rather advanced tool that provides a lot of different modes and was used as part of the feature extraction in this thesis. IntaRNA in its simplest mode can be used to scan genomes for target sites of specific small RNAs using a heuristics based approach that can be modified via a different number of command line arguments. The next mode in terms of simplicity is called IntaRNA exact and it aims to minimize the following sum:

$$E_{hybrid}(I) + ED1(I) + ED2(I) \tag{1.1}$$

In this model, E_{hybrid} denotes the hybridization energy of the base pairs forming between the two distinct molecules. Meanwhile, E_D represents the accessibility energy, the energy required to break existing internal (intra-molecular) bonds so the target sequences can interact. The model evaluates specific inter-molecular binding patterns, such as helical stackings, bulges, and interior loops. Because internal base pairings are not strictly mapped out, IntaRNA's predictions are flexible enough to account for any potential structural arrangement during single-site interactions. The ensemble mode of IntaRNA is based on a partition function computation that aims to find the optimal target site among many on the target sequence such that:

$$\arg\max Z(S) = \exp\left(-\frac{E_{D1}(S)}{RT}\right) \exp\left(-\frac{E_{D2}(S)}{RT}\right) \sum \exp\left(-\frac{E_{hybrid}(I)}{RT}\right) \tag{1.2}$$

The site-specific partition function, $Z(S)$, is calculated by multiplying the Boltzmann weights of the accessibility penalties (E_{D1} and E_{D2}) by the sum of the Boltzmann weights for all hybridization interactions (I) at that site. Here, R represents the universal gas constant and T is the temperature (with $RT \approx 0.616$ at 37°C). This partition function is used to determine the ensemble energy of all interactions at a given site using the equation $E(S) = -RT \ln(Z(S))$. By calculating ensemble energy rather than relying on a single, static base-pairing model, this approach accounts for the inherent flexibility and dynamic nature of the interacting regions. Furthermore, summing the partition functions across all potential sites yields the global partition function, $Z_{all} = \sum Z(S)$. This allows for the calculation of the probability that a specific interaction I will form, defined as $P_E(I) = \exp(-E(I)/RT)/Z_{all}$. IntaRNA provides additional options that function like other widely used software for the same task like RNADuplex, TargetScan and RNAup (Lorenz, Bernhart, *et al.* 2011; Hofacker *et al.* 1994; Lorenz, Hofacker, *et al.* 2016; Turner & Mathews 2010).

1.2.2 Functional and Pathway Level predictors

The advancement of machine and deep learning algorithms has paved the way for the development of models that circumvent the need for feature engineering and go beyond site level prediction. Heuristic and Physical approaches as those described above, as well as classical machine learning algorithms, need to take as input a number of features among which they try to calculate weights and correlations so as to create a function that separates the samples into predifined categories or fit a regression line on a multi-dimensional distribution. This requires prior processing of the samples so as to extract such meaningful features utilizing domain knowledge for the specific task. Modern deep learning models do not require such intensive preprocessing as they are developed in order to process the nucleotide sequence directly and find more abstract patterns that correlate with the desired output. The first and simplest type of architecture that has been used is Convolutional Neural Networks (CNNs). CNNs allow for the processing of the nucleotide sequence outside of the tabular limitations of classical Machine Learning by reading it in predifined windows along which mathematical operations are performed with the goal of creating a feature map whereby pooling is performed in order to reduce overfitting and make the model translation invariant. The final part usually consists of a classification layer that is used to output a probability for a specific class.

Some modern architectures like the one from Zacharopoulou *et al.* 2024 employ such architectures combining them with ones like Graph Neural Networks (GNNs) in order to extract a unified score across an entire gene and use it as a feature for a meta learner that is trained on expression data to predict functional binding which goes beyond site level predictions. The most recent and advanced models perform what is called Pathway Level prediction, by taking advantadge of site/gene level predictors and GNNs, they predict pathway level interactions between mRNAs-miRNAs-Competing Endogenous RNAs (ceRNAs), making them more relevant for identifying which of these pathways are crucial for disease states (Chiu *et al.* 2015; Jin *et al.* 2024). Finally, modern models may also rely on Transformer based architectures which are able to treat DNA/RNA as a language and learn more intricate pattern that are not necessarily in proximity in the input sequence. Examples of these include the vanilla Transformer, Titans and the Mamba architecture which is not transformer based but has shown promise in DNA related tasks (Behrouz *et al.* 2024; Gu & Dao 2024)

1.2.3 Transformers

Transformers were introduced in the landmark paper ”Attention is All You Need” (Vaswani *et al.* 2023). First, the input sequence is tokenized and mapped to dense vector embeddings. To inject information about the relative or absolute position of the tokens in the sequence, positional encodings, which are either learned or based on sinusoidal functions, are added to these embeddings. Through learned linear projections, the input is mapped into three distinct matrices: Queries (Q), Keys (K), and Values (V). The scaled dot-product attention mechanism, which is vital to the architecture’s success, is defined

by the following formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1.3}$$

Rather than computing a single attention function, Transformers employ Multi-Head Attention, projecting the queries, keys, and values into multiple subspaces, performing the attention function in parallel, and concatenating the outputs. The standard Transformer encoder consists of a stack of identical layers. Each layer contains two primary sub-layers: a Multi-Head Attention mechanism and a position-wise Feed-Forward Neural Network. Every sub-layer is encompassed by a residual connection followed by layer normalization, formulated mathematically as LayerNorm$(x + \text{Sublayer}(x))$.

The Decoder follows a similar structural philosophy to the Encoder but introduces critical modifications to facilitate autoregressive sequence generation. Each identical decoder layer comprises three primary sub-layers: a Masked Multi-Head Self-Attention mechanism, an Encoder-Decoder Multi-Head Attention mechanism (often termed Cross-Attention), and a position-wise Feed-Forward Neural Network. As with the Encoder, residual connections and layer normalization encompass each of these sub-layers.

The masking in the first sub-layer is an essential modification; it mathematically prevents sequence positions from attending to subsequent positions. By setting the scaled dot-product attention scores of future tokens to $-\infty$ prior to the softmax operation, the model ensures that the prediction for position i depends exclusively on the known outputs at positions strictly prior to i.

In the subsequent Encoder-Decoder Attention sub-layer, the queries (Q) are derived from the output of the previous decoder layer, while the keys (K) and values (V) are sourced directly from the final output of the Encoder. This cross-attention mechanism allows every position in the decoder to attend over the entire input sequence, effectively grounding the generated output in the original provided context. Finally, the output of the decoder stack is passed through a linear transformation and a softmax function to produce a probability distribution over the vocabulary for the next predicted token.

1.2.4 Mamba

An alternative and highly successful paradigm has emerged in the form of Structured State Space Sequence Models (SSMs), most notably the Mamba architecture (Gu & Dao 2024). Mamba completely discards the attention mechanism, offering a fundamental architectural shift that scales linearly in time ($\mathcal{O}(N)$) with respect to sequence length, while maintaining a constant memory footprint during autoregressive generation.

Mathematical Foundation of Structured State Space Models

At its core, Mamba is inspired by continuous-time control theory. It maps a 1-dimensional input sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an implicit latent state $h(t) \in \mathbb{R}^N$.

The classical continuous-time SSM is defined by a system of linear ordinary differential equations:

$$h'(t) = Ah(t) + Bx(t) \tag{1.4}$$

$$y(t) = Ch(t) \tag{1.5}$$

To process discrete tokens (such as text), this continuous system must be discretized. Using a timescale parameter Δ, the continuous matrices A and B are transformed into their discrete counterparts $\bar{A}$ and $\bar{B}$ (typically via the zero-order hold method). This yields a highly efficient recurrent formulation:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t \tag{1.6}$$

Previous iterations of SSMs (such as S4) suffered from a fatal constraint: they were Linear Time-Invariant (LTI). Because their transition matrices (B, C, and Δ) were static and fixed across all time steps, they could not selectively focus on or ignore specific tokens, making tasks like targeted information retrieval or copy-pasting nearly impossible.

Mamba resolves this by introducing a *Selection Mechanism*. It parameterizes B, C, and the step size Δ to be input-dependent functions of the current token x_t:

$$B_t = \text{Linear}_B(x_t), \quad C_t = \text{Linear}_C(x_t), \quad \Delta_t = \text{softplus}(\text{Linear}_\Delta(x_t)) \tag{1.7}$$

By allowing Δ_t to vary based on the input, the model dynamically dictates the degree to which it should integrate the current token into the hidden state (h_t) or completely ignore it. This effectively grants the recurrent model the data-filtering capabilities of an attention mechanism, without the $\mathcal{O}(N^2)$ computational penalty.

Because Mamba's parameters are time-varying, the model cannot be computed using efficient Fast Fourier Transforms (FFTs) like its LTI predecessors. To overcome this, Mamba relies on a hardware-aware parallel scan algorithm. By fusing the scan operations and materializing the expanded hidden states (h_t) exclusively in the faster, on-chip SRAM of the GPU (bypassing the slower High Bandwidth Memory (HBM) IO bottleneck) Mamba achieves training speeds competitive with highly optimized Transformers while preserving the inference efficiency of a recurrent neural network.

1.2.5 Main Goal

The main goal of this thesis is to explore the use of architectures and algorithms that aim to surpass Transformers in terms of either performance or explainability. Thus,the Mamba architecture was employed and its results were compared to those of a vanilla Transformer deep neural network. Additionally, simpler machine learning models and their ensembles were employed in order to further examine the tradeoff between their higher degree of explainability and (potentially) speed and their performance on the Manakov *et al.* 2022 dataset.

2. METHODS

2.1 Data preprocessing

The dataset used for training and testing was taken from Sammut *et al.* 2025 and contains preprocessed files from experimental results produced by Manakov *et al.* 2022; Hejret *et al.* 2023. These include, apart from the DNA sequences of the miRNA and the MREs, information about the miRNA family, the isomiR that the molecule is identified as, the hg38 genomic coordinates of the MRE, two types of conservation vector scores (genePhastCons100way and genePhyloP) in lists, what region of the MRE is targeted (exon, intron, 3'UTR, 5'UTR and boundaries between them), gene cluster ID (which is used for negative sample generation) and labels (1 for positives, 0 for negatives). The dataset is balanced in terms of positives and negatives not only in its totality but also within miRNA families. This is crucial, as different families may bind in different manners and with different criteria and thus it is important that any potential model learns all such patterns.

The steps for negative sample generation according to Sammut *et al.* 2025 are as follows:

1. Cluster MREs of >90% similarity
2. For a given miRNA family:
   (a) Identify clusters containing targets of that family
   (b) Choose an MRE from a cluster that does not overlap with the aforementioned ones
   (c) Assign it to each miRNA of that family to create the negative sample pair

Another bias that occurs apart from the skewed positives to negative ratio, has to do with the tissue of origin. More specifically, given that the dataset used comes from a single cell line (in this case HEK-293), it is essentially certain that there will be a sampling bias since tissue specific expression is not uniform. Therefore, specific miRNAs and targets are expressed more than others and are more likely to be captured by the experimental procedure. Adding to that, specific interactions of miRNA and MREs might exist solely due to this tissue specific expression pattern and conversely, specific negative samples might be detected as valid interactions if one was to use another tissue for the same experiment, due to different expression patterns. In order to mitigate this, data from TarBase (SETHUPATHY *et al.* 2006) were extracted in order to assess which of the interactions in the Manakov dataset were truly negative, based on their absence in experimental data, and which are false negatives that could hinder the models ability to learn binding patterns.

The latest version of the file TarBase provides for Homo sapiens was downloaded (v9) and cleaning was performed so as to keep only valid entries with existing genomic coordinates. If an entry, that is detected as a pair of miRNA name - Gene name, had missing coordinates for the target site it was either dropped (if the same pair exists either within the same trascript region or not) or the coordinates from the entire gene were kept from Ensembl (in the case no other entry for that pair existed). The goal of this process is to keep as many valid pairs of miRNA - Target gene from TarBase along with coordinates that can be used for overlapping with the ones from the dataset later.

The pipeline to remove falsely assigned negative samples from the dataset adheres to the steps below in order to retain the initial positive to negative ratio:

For a given miRNA:

1. Check for overlapping coordinates of its target between the cleaned TarBase file and the negative samples of the dataset
2. If overlaps are found, remove the corresponding negative sample and keep track of removed samples per family
3. Remove at random as many positive samples from each family

Another aspect of the sampling bias that had to be adressed was the fact that the model has to be prevented from treating samples from rarer families as less important and therefore ignoring potential novel binding rules in the process. Sampling weights were added to the datasets that adhere to the following formula:

$$\text{Sample Weight} = \frac{\text{Sample Total}}{\max(100, \text{Family Total} \times \text{Count of current Family})} \tag{2.1}$$

The above formula ensures that samples of more abundant miRNA families get assigned with a smaller weight proportional to their family's frequency in the dataset, while rarer samples are assigned bigger weights capped based on the product in the denominator. It is worth noting that when this ratio is multiplied by the current family count, the result is equal to the same number, with small deviations for extremely rare families.

Since the goal of the model is to be able to generalize its predictions for unseen data and more specifically unseen families, data are split based on a stratified kfold cross validation scheme that takes into account the miRNA family for each split, ensuring that there are no common miRNA families across splits and that the ratio of positives and negatives is maintained. In that manner the model of each fold is trained on specific miRNA families while the ones in the validation set are used for tuning it. Furthermore, a separate test set exists that has unseen samples of families that are common with the training set along with a holdout set with completely unseen families to both the training and test set as well as to the other two files contained in the study. Every model is tested on both of these test sets in the end of its training in order to evaluate its performance. The main metric that is used for this purpose is the Average Precision score: APS $= \sum_n (R_n - R_{n-1}) P_n$ where

R_n and P_n are the Recall and the Precision at the nth threshold. APS is used because it is threshold invariant and allows for the evaluation of the model's capability to find the positive class. This is important as validation is costly and thus False Positive samples and False Negative ones must be penalized, while True Negative samples are not equally meaningful.

3. RESULTS

3.1 Language Models

Inspired by microT-CNN (Zacharopoulou *et al.* 2024), two approaches were followed with the final goal of making an enhanced algorithm for achieving site-level miRNA target site prediction. Language model architectrures were developed based on Mamba in order to test their capabilities of learning miRNA binding rules based only on nucleotide sequences. The initial experiment involved using the kmer encoded (k=4) chimeric sequences, that are the result of merging the miRNA and MRE sequences as input to a Transformer model. Unfortunately, this led to a model that seemed to be underfitting despite subsequent trials to fine tune its hyperparameters. The same held true even with the attempt of making a siamese neural network that processed the miRNA and the MRE sequences separately and tried to minimize the distance between them in the embedded space. The Mamba architecture seemed to yield similar results and thus a change in terms of encoding was needed.

The kmer encoding was swapped with One Hot encoding and

3.2 Machine Learning

Starting with

4. FIGURES AND TABLES

4.1 Inserting tables

Below is a sample table with its caption. Make sure you put the caption **over** the table. For instance, this code in LaTeX:

```
\begin{table}
    % this is before the table
    \caption{A sample event log, sorted by timestamp}
    % this is to label and include it in our list of tables
    \label{tbl:event_log_sample}
    \begin{center} % center the table
    \begin{tabular}{c|c|c}
        \textbf{Case Id} & \textbf{Activity} & \textbf{Timestamp} \\
        \hline
        1 & Activity A & 2021-05-24T12:30:00Z \\
        1 & Activity B & 2021-05-24T12:30:26Z \\
        1 & Activity C & 2021-05-24T12:31:04Z \\
        2 & Activity A & 2021-05-24T13:19:39Z \\
        2 & Activity B & 2021-05-24T13:19:50Z \\
        3 & Activity A & 2021-05-24T13:19:59Z \\
        2 & Activity C & 2021-05-24T13:22:14Z \\
        3 & Activity B & 2021-05-24T13:24:24Z \\
        4 & Activity A & 2021-05-24T13:25:06Z \\
        3 & Activity C & 2021-05-24T13:25:27Z \\
        4 & Activity B & 2021-05-24T13:25:53Z \\
        4 & Activity C & 2021-05-24T13:27:48Z \\
        \hline
    \end{tabular}
    \end{center}
\end{table}
```

renders table 4.1:

Table 4.1: A sample event log, sorted by timestamp

Case Id	Activity	Timestamp
1	Activity A	2021-05-24T12:30:00Z
1	Activity B	2021-05-24T12:30:26Z
1	Activity C	2021-05-24T12:31:04Z
2	Activity A	2021-05-24T13:19:39Z
2	Activity B	2021-05-24T13:19:50Z
3	Activity A	2021-05-24T13:19:59Z
2	Activity C	2021-05-24T13:22:14Z
3	Activity B	2021-05-24T13:24:24Z
4	Activity A	2021-05-24T13:25:06Z
3	Activity C	2021-05-24T13:25:27Z
4	Activity B	2021-05-24T13:25:53Z
4	Activity C	2021-05-24T13:27:48Z

4.2 Inserting figures

Below is an example of inserting a figure with its caption. Make sure you put the caption **under** the figure. For example, this LaTeX code:

ABBREVIATIONS - ACRONYMS

AI	Artificial Intelligence
SPARQL	SPARQL Protocol and RDF Query Language
OWL	Web Ontology Language
OGC	Open Geospatial Consortium

APPENDIX A. FIRST APPENDIX

APPENDIX B. SECOND APPENDIX

REFERENCES

1. A, H., G, K., T, D. & D, T. Mapping the Human miRNA Interactome by CLASH Reveals Frequent Noncanonical Binding. *Cell* **153.** ISSN: 1097-4172. PMID: 23622248. https://pubmed.ncbi.nlm.nih.gov/23622248/?dopt=Abstract (2026) (Apr. 25, 2013).
2. Adam, L. *et al.* miR-200 Expression Regulates Epithelial-to-Mesenchymal Transition in Bladder Cancer Cells and Reverses Resistance to Epidermal Growth Factor Receptor Therapy. *Clinical Cancer Research* **15,** 5060–5072. ISSN: 1078-0432. https://doi.org/10.1158/1078-0432.CCR-08-2245 (2026) (Aug. 12, 2009).
3. Addo-Quaye, C., Eshoo, T. W., Bartel, D. P. & Axtell, M. J. Endogenous siRNA and miRNA Targets Identified by Sequencing of the Arabidopsis Degradome. *Current Biology* **18,** 758–762. ISSN: 0960-9822. PMID: 18472421. https://www.cell.com/current-biology/abstract/S0960-9822(08)00528-9 (2026) (May 20, 2008).
4. AMBROS, V. I. C. T. O. R. *et al.* A Uniform System for microRNA Annotation. *RNA* **9,** 277–279. ISSN: 1355-8382. PMID: 12592000. https://pmc.ncbi.nlm.nih.gov/articles/PMC1370393/ (2026) (Mar. 2003).
5. Behrouz, A., Zhong, P. & Mirrokni, V. *Titans: Learning to Memorize at Test Time* version 1. arXiv: 2501.00663 [cs]. http://arxiv.org/abs/2501.00663 (2025). Prepublished.
6. Benesova, S., Kubista, M. & Valihrach, L. Small RNA-Sequencing: Approaches and Considerations for miRNA Analysis. *Diagnostics* **11,** 964. ISSN: 2075-4418. PMID: 34071824. https://pmc.ncbi.nlm.nih.gov/articles/PMC8229417/ (2026) (May 27, 2021).
7. Busch, A., Richter, A. S. & Backofen, R. IntaRNA: Efficient Prediction of Bacterial sRNA Targets Incorporating Target Site Accessibility and Seed Regions. *Bioinformatics* **24,** 2849–2856. ISSN: 1367-4811. PMID: 18940824 (Dec. 15, 2008).
8. Chaluvally-Raghavan, P. *et al.* Direct Upregulation of STAT3 by MicroRNA-551b-3p Deregulates Growth and Metastasis of Ovarian Cancer. *Cell Reports* **15,** 1493–1504. ISSN: 2211-1247. PMID: 27160903 (May 17, 2016).
9. Chiu, H.-S. *et al.* Cupid: Simultaneous Reconstruction of microRNA-target and ceRNA Networks. *Genome Research* **25,** 257–267. ISSN: 1088-9051. PMID: 25378249. https://pmc.ncbi.nlm.nih.gov/articles/PMC4315299/ (2026) (Feb. 2015).
10. Corbin, R., Olsson-Carter, K. & Slack, F. The Role of microRNAs in Synaptic Development and Function. *BMB reports* **42,** 131–135. ISSN: 1976-6696. PMID: 19335998. https://pmc.ncbi.nlm.nih.gov/articles/PMC4315497/ (2026) (Mar. 31, 2009).
11. Diener, C., Keller, A. & Meese, E. The miRNA–Target Interactions: An Underestimated Intricacy. *Nucleic Acids Research* **52,** 1544–1557. ISSN: 0305-1048. https://doi.org/10.1093/nar/gkad1142 (2026) (Feb. 28, 2024).

12. Ding, S., Huang, H., Xu, Y., Zhu, H. & Zhong, C. MiR-222 in Cardiovascular Diseases: Physiology and Pathology. *BioMed Research International* **2017,** 4962426. ISSN: 2314-6133. PMID: 28127557. https://pmc.ncbi.nlm.nih.gov/articles/PMC5239839/ (2026) (2017).

13. Gelhausen, R., Will, S., Hofacker, I. L., Backofen, R. & Raden, M. IntaRNAhelix-composing RNA-RNA Interactions from Stable Inter-Molecular Helices Boosts Bacterial sRNA Target Prediction. *Journal of Bioinformatics and Computational Biology* **17,** 1940009. ISSN: 1757-6334. PMID: 31856671 (Oct. 2019).

14. German, M. A., Luo, S., Schroth, G., Meyers, B. C. & Green, P. J. Construction of Parallel Analysis of RNA Ends (PARE) Libraries for the Study of Cleaved miRNA Targets and the RNA Degradome. *Nature Protocols* **4,** 356–362. ISSN: 1750-2799. PMID: 19247285 (2009).

15. Gregory, B. D. *et al.* A Link between RNA Metabolism and Silencing Affecting Arabidopsis Development. *Developmental Cell* **14,** 854–866. ISSN: 1878-1551. PMID: 18486559 (June 2008).

16. Grimson, A. *et al.* MicroRNA Targeting Specificity in Mammals: Determinants beyond Seed Pairing. *Molecular Cell* **27,** 91–105. ISSN: 1097-2765. PMID: 17612493. https://www.cell.com/molecular-cell/abstract/S1097-2765(07)00407-8 (2026) (July 6, 2007).

17. Gu, A. & Dao, T. *Mamba: Linear-Time Sequence Modeling with Selective State Spaces* arXiv: 2312.00752 [cs]. http://arxiv.org/abs/2312.00752 (2025). Prepublished.

18. Güller, I. & Russell, A. P. MicroRNAs in Skeletal Muscle: Their Role and Regulation in Development, Disease and Function. *The Journal of Physiology* **588,** 4075–4087. ISSN: 0022-3751. PMID: 20724363. https://pmc.ncbi.nlm.nih.gov/articles/PMC3002442/ (2026) (Pt 21 Nov. 1, 2010).

19. Guo, H. *et al.* Nuclear miR-30b-5p Suppresses TFEB-mediated Lysosomal Biogenesis and Autophagy. *Cell Death and Differentiation* **28,** 320–336. ISSN: 1476-5403. PMID: 32764647 (Jan. 2021).

20. Hejret, V. *et al.* Analysis of Chimeric Reads Characterises the Diverse Targetome of AGO2-mediated Regulation. *Scientific Reports* **13,** 22895. ISSN: 2045-2322. https://www.nature.com/articles/s41598-023-49757-z (2025) (Dec. 21, 2023).

21. Hofacker, I. L. *et al.* Fast Folding and Comparison of RNA Secondary Structures. *Monatshefte für Chemie / Chemical Monthly* **125,** 167–188. ISSN: 1434-4475. https://doi.org/10.1007/BF00818163 (2026) (Feb. 1, 1994).

22. Hornstein, E. *et al.* The microRNA miR-196 Acts Upstream of Hoxb8 and Shh in Limb Development. *Nature* **438,** 671–674. ISSN: 1476-4687. https://www.nature.com/articles/nature04138 (2026) (Dec. 2005).

23. Jin, Z., Shi, Y. & Zhou, L. Transparent Sparse Graph Pathway Network for Analyzing the Internal Relationship of Lung Cancer. *Frontiers in Genetics* **15.** ISSN: 1664-8021. https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2024.1437174/full (2026) (Oct. 1, 2024).

24. Kim, D. H., Saetrom, P., Snøve, O. & Rossi, J. J. MicroRNA-directed Transcriptional Gene Silencing in Mammalian Cells. *Proceedings of the National Academy of Sciences of the United States of America* **105,** 16230–16235. ISSN: 1091-6490. PMID: 18852463 (Oct. 21, 2008).

25. Kozomara, A. & Griffiths-Jones, S. miRBase: Integrating microRNA Annotation and Deep-Sequencing Data. *Nucleic Acids Research* **39,** D152–D157. ISSN: 0305-1048. PMID: 21037258. https://pmc.ncbi.nlm.nih.gov/articles/PMC3013655/ (2026) (Database issue Jan. 2011).

26. Langmead, B., Trapnell, C., Pop, M. & Salzberg, S. L. Ultrafast and Memory-Efficient Alignment of Short DNA Sequences to the Human Genome. *Genome Biology* **10,** R25. ISSN: 1474-760X. https://doi.org/10.1186/gb-2009-10-3-r25 (2025) (Mar. 4, 2009).

27. Lewis, B. P., Burge, C. B. & Bartel, D. P. Conserved Seed Pairing, Often Flanked by Adenosines, Indicates That Thousands of Human Genes Are microRNA Targets. *Cell* **120,** 15–20. ISSN: 0092-8674. PMID: 15652477 (Jan. 14, 2005).

28. Lewis, B. P., Shih, I.-h., Jones-Rhoades, M. W., Bartel, D. P. & Burge, C. B. Prediction of Mammalian MicroRNA Targets. *Cell* **115,** 787–798. ISSN: 0092-8674, 1097-4172. PMID: 14697198. https://www.cell.com/cell/abstract/S0092-8674(03)01018-3 (2026) (Dec. 26, 2003).

29. Li, H. *Aligning Sequence Reads, Clone Sequences and Assembly Contigs with BWA-MEM* arXiv: 1303.3997 [q-bio]. http://arxiv.org/abs/1303.3997 (2026). Pre-published.

30. Lin, F. *et al.* Bladder Cancer Cell-Secreted Exosomal miR-21 Activates the PI3K/AKT Pathway in Macrophages to Promote Cancer Progression. *International Journal of Oncology* **56,** 151–164. ISSN: 1019-6439. PMID: 31814034. https://pmc.ncbi.nlm.nih.gov/articles/PMC6910194/ (2026) (Dec. 3, 2019).

31. Lorenz, R., Bernhart, S. H., *et al.* ViennaRNA Package 2.0. *Algorithms for Molecular Biology* **6,** 26. ISSN: 1748-7188. https://doi.org/10.1186/1748-7188-6-26 (2026) (Nov. 24, 2011).

32. Lorenz, R., Hofacker, I. L. & Stadler, P. F. RNA Folding with Hard and Soft Constraints. *Algorithms for molecular biology: AMB* **11,** 8. ISSN: 1748-7188. PMID: 27110276 (2016).

33. Manakov, S. A. *et al. Scalable and Deep Profiling of mRNA Targets for Individual microRNAs with Chimeric eCLIP* https://www.biorxiv.org/content/10.1101/2022.02.13.480296v2 (2026). Pre-published.

34. Mann, M., Wright, P. R. & Backofen, R. IntaRNA 2.0: Enhanced and Customizable Prediction of RNA-RNA Interactions. *Nucleic Acids Research* **45,** W435–W439. ISSN: 1362-4962. PMID: 28472523 (July 3, 2017).

35. Miladi, M., Montaseri, S., Backofen, R. & Raden, M. Integration of Accessibility Data from Structure Probing into RNA–RNA Interaction Prediction. *Bioinformatics* **35,** 2862–2864. ISSN: 1367-4803. https://doi.org/10.1093/bioinformatics/bty1029 (2026) (Aug. 15, 2019).

36. O'Brien, J., Hayder, H., Zayed, Y. & Peng, C. Overview of MicroRNA Biogenesis, Mechanisms of Actions, and Circulation. *Frontiers in Endocrinology* **9,** 402. ISSN: 1664-2392. PMID: 30123182. https://pmc.ncbi.nlm.nih.gov/articles/PMC6085463/ (2026) (Aug. 3, 2018).

37. Raden, M., Müller, T., Mautner, S., Gelhausen, R. & Backofen, R. The Impact of Various Seed, Accessibility and Interaction Constraints on sRNA Target Prediction-a Systematic Assessment. *BMC Bioinformatics* **21,** 15. ISSN: 1471-2105. https://doi.org/10.1186/s12859-019-3143-4 (2026) (Jan. 13, 2020).

38. Sammut, S. *et al.* miRBench: Novel Benchmark Datasets for microRNA Binding Site Prediction That Mitigate against Prevalent microRNA Frequency Class Bias. *Bioinformatics* **41,** i542–i551. ISSN: 1367-4811. https://doi.org/10.1093/bioinformatics/btaf233 (2026) (Supplement_1 July 1, 2025).

39. Sapre, N. *et al.* A Urinary microRNA Signature Can Predict the Presence of Bladder Urothelial Carcinoma in Patients Undergoing Surveillance. *British Journal of Cancer* **114,** 454–462. ISSN: 1532-1827. https://www.nature.com/articles/bjc2015472 (2026) (Feb. 2016).

40. SETHUPATHY, P. R. A. V. E. E. N., CORDA, B. E. N. O. I. T. & HATZIGEORGIOU, A. G. TarBase: A Comprehensive Database of Experimentally Supported Animal microRNA Targets. *RNA* **12,** 192–197. ISSN: 1355-8382. PMID: 16373484. https://pmc.ncbi.nlm.nih.gov/articles/PMC1370898/ (2026) (Feb. 2006).

41. Tomasello, L., Distefano, R., Nigita, G. & Croce, C. M. The MicroRNA Family Gets Wider: The IsomiRs Classification and Role. *Frontiers in Cell and Developmental Biology* **9,** 668648. ISSN: 2296-634X. PMID: 34178993. https://pmc.ncbi.nlm.nih.gov/articles/PMC8220208/ (2026) (June 9, 2021).

42. Travis, A. J., Moody, J., Helwak, A., Tollervey, D. & Kudla, G. Hyb: A Bioinformatics Pipeline for the Analysis of CLASH (Crosslinking, Ligation and Sequencing of Hybrids) Data. *Methods (San Diego, Calif.)* **65,** 263–273. ISSN: 1046-2023. PMID: 24211736. https://pmc.ncbi.nlm.nih.gov/articles/PMC3969109/ (2026) (Feb. 2014).

43. Turner, D. H. & Mathews, D. H. NNDB: The Nearest Neighbor Parameter Database for Predicting Stability of Nucleic Acid Secondary Structure. *Nucleic Acids Research* **38,** D280–282. ISSN: 1362-4962. PMID: 19880381 (Database issue Jan. 2010).

44. Vaswani, A. *et al. Attention Is All You Need* arXiv: 1706.03762 [cs]. http://arxiv.org/abs/1706.03762 (2026). Pre-published.

45. Wright, P. R. *et al.* CopraRNA and IntaRNA: Predicting Small RNA Targets, Networks and Interaction Domains. *Nucleic Acids Research* **42,** W119–123. ISSN: 1362-4962. PMID: 24838564 (Web Server issue July 2014).

46. Xiao, M. *et al.* MicroRNAs Activate Gene Transcription Epigenetically as an Enhancer Trigger. *RNA Biology* **14,** 1326–1334. ISSN: 1547-6286. PMID: 26853707. https://doi.org/10.1080/15476286.2015.1112487 (2026) (Oct. 3, 2017).

47. Zacharopoulou, E. *et al.* microT-CNN: An Avant-Garde Deep Convolutional Neural Network Unravels Functional miRNA Targets beyond Canonical Sites. *Briefings in Bioinformatics* **26,** bbae678. ISSN: 1467-5463. PMID: 39737571. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11685103/ (2025) (Dec. 31, 2024).

48. Zeng, Y. & Cullen, B. R. Sequence Requirements for Micro RNA Processing and Function in Human Cells. *RNA* **9,** 112–123. ISSN: 1355-8382, 1469-9001. PMID: 12554881. http://rnajournal.cshlp.org/content/9/1/112 (2026) (Jan. 1, 2003).

49. Zhang, X. & Zeng, Y. The Terminal Loop Region Controls microRNA Processing by Drosha and Dicer. *Nucleic Acids Research* **38,** 7689–7697. ISSN: 0305-1048. https://doi.org/10.1093/nar/gkq645 (2026) (Nov. 1, 2010).